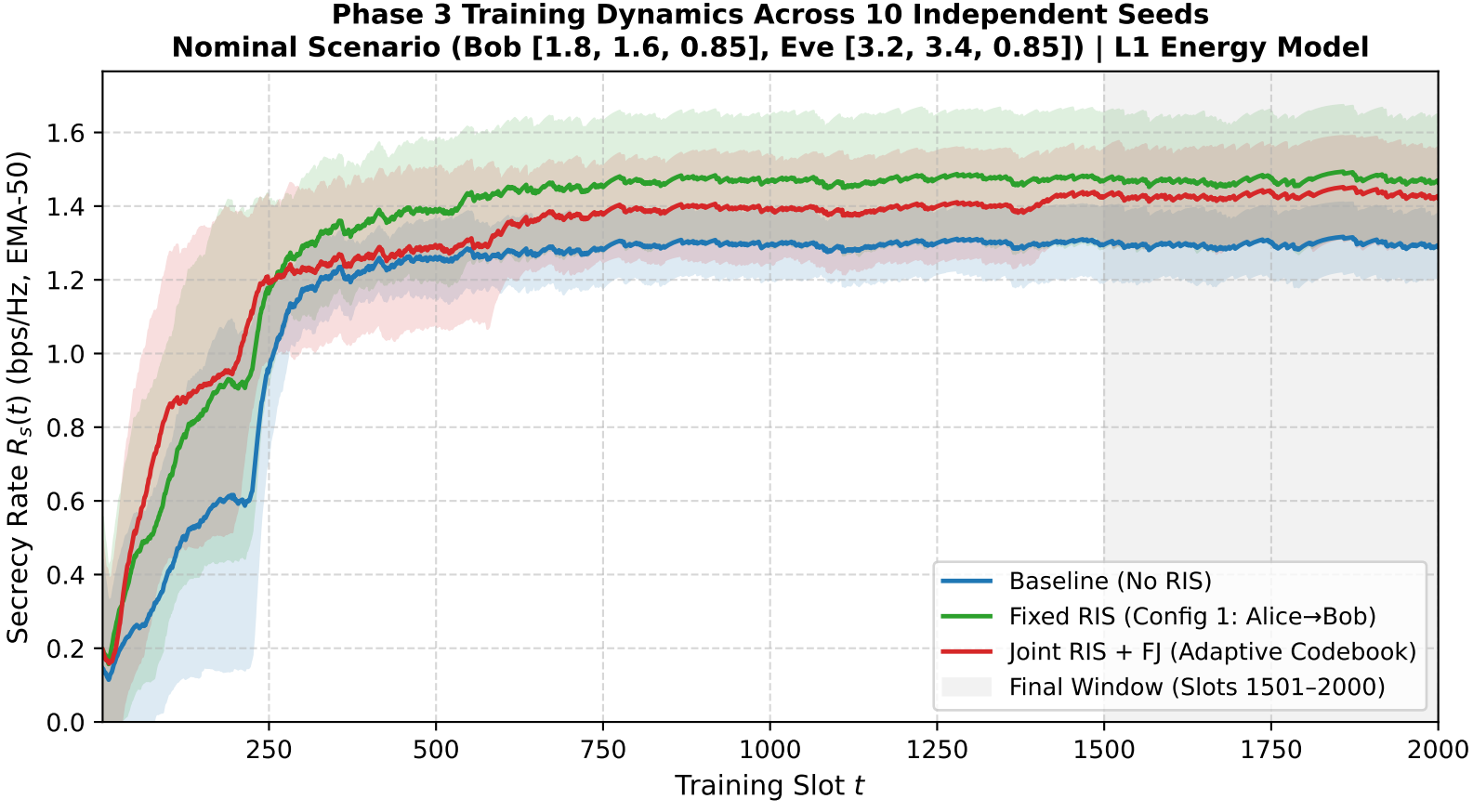

Phase 3 Training Dynamics Across 10 Independent Seeds
Nominal Scenario (Bob [1.8, 1.6, 0.85], Eve [3.2, 3.4, 0.85]) | L1 Energy Model
Secrecy Rate $R_s(t)$ (bps/Hz, EMA-50)
Training Slot t
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
250
500
750
1000
1250
1500
1750
2000
Baseline (No RIS)
Fixed RIS (Config 1: Alice→Bob)
Joint RIS + FJ (Adaptive Codebook)
Final Window (Slots 1501–2000)